

Can the Density Operator Be Determined from One Observable?

Introduction

In quantum mechanics, the density operator $\hat{\rho}$ fully characterizes the state of a system, including both pure and mixed states. A natural question is:

Can the density operator be determined by measuring only one observable?

The answer is **no**. Below we explain why, with theory and examples.

1. What information does one observable give?

Let $\hat{A}$ be an observable with spectral decomposition

$$\hat{A} = \sum_n a_n |a_n\rangle \langle a_n|. \quad (1)$$

If the system is in state $\hat{\rho}$, the probability of outcome a_n is

$$P(a_n) = \text{Tr}(\hat{\rho} |a_n\rangle \langle a_n|). \quad (2)$$

These probabilities determine only the diagonal elements of $\hat{\rho}$ in the $\{|a_n\rangle\}$ basis:

$$\rho_{nn} = \langle a_n | \hat{\rho} | a_n \rangle. \quad (3)$$

The off-diagonal elements remain completely undetermined.

2. Counting argument

For a Hilbert space of dimension d :

- A density matrix has $d^2 - 1$ real parameters.
- One observable yields only $d - 1$ independent probabilities.

Thus, one observable cannot determine $\hat{\rho}$ uniquely.

3. Example: spin-1/2 system

Let

$$\hat{\rho} = \begin{pmatrix} \rho_{11} & \rho_{12} \\ \rho_{12}^* & \rho_{22} \end{pmatrix}, \quad \rho_{11} + \rho_{22} = 1. \quad (4)$$

Measure only

$$\hat{S}_z = \frac{\hbar}{2} \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}. \quad (5)$$

Then

$$P(+\hbar/2) = \rho_{11}, \quad (6)$$

$$P(-\hbar/2) = \rho_{22}. \quad (7)$$

The coherence ρ_{12} is completely inaccessible.

Thus, states such as

$$\hat{\rho}_1 = \begin{pmatrix} 1/2 & 0 \\ 0 & 1/2 \end{pmatrix}, \quad \hat{\rho}_2 = \begin{pmatrix} 1/2 & 1/2 \\ 1/2 & 1/2 \end{pmatrix} \quad (8)$$

give identical measurement results for S_z , but represent different physical states.

4. What is required instead?

To determine $\hat{\rho}$, one must perform **quantum state tomography**, which requires:

- Measuring several non-commuting observables,
- Collecting enough data to reconstruct all $d^2 - 1$ parameters.

For a qubit, this means measuring S_x , S_y , and S_z .

Conclusion

A single observable is insufficient to determine the density operator. Only diagonal elements in that observable's eigenbasis can be found. Full knowledge of the quantum state requires multiple, non-commuting measurements.